

Progress Test 2.B — Thermo-S1

2.B · Thermoelastic behaviour of gases
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

A gas has a piezothermic coefficient $\beta = 3.66 \times 10^{-3} \text{ K}^{-1}$. Express β in $^{\circ}\text{C}^{-1}$.

[Empty grid for Exercise 1]

Marking

①
 0 1
 [] []

Exercise 2. Piezothermic coefficient — definition (5 marks)

Define the piezothermic coefficient β of a gas. Give its formula, the physical variation it measures, its sign and its SI unit. How does it differ from α ?

[Empty grid for Exercise 2]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Gases at constant volume

(3 marks)

A gas enclosed in a rigid, sealed container is heated. Its pressure:

- ☐ increases ☐ decreases ☐ does not change ☐ becomes zero

The coefficient β measures the variation of:

- ☐ V at fixed p ☐ p at fixed V
☐ T at fixed V ☐ V at fixed T

For a car tyre (airtight, rigid rim), the variables held fixed are:

- ☐ n and V ☐ n and p ☐ T and V ☐ p and V

Exercise 4. Tyre pressure

(5 marks)

A car tyre, whose volume is fixed, is inflated to p_1 at T_1 . While driving, its temperature rises by ΔT . Using the value of β given below (assumed constant), estimate the pressure increase Δp and the final pressure p_2 .

$p_1 = 2.0 \text{ bar}$	$T_1 = 300 \text{ K}$	$\Delta T = 30 \text{ K}$	$\beta = \frac{1}{300} \text{ K}^{-1}$
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Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

— End of the test —